

GRADE 10 MATHEMATICS NOVEMBER FINAL EXAM PAPER 1 (ALGEBRA, FUNCTIONS & FINANCE)

Subject: Mathematics | Grade: 10 | Academic Year: 2024 | Assessment: Term 4 Examination

Document Type: Standard Examination Question Paper • Time: 2 - 3 Hours • Total Marks: 100 - 150

INSTRUCTIONS AND INFORMATION

- This document consists of all standard curriculum questions and study guidelines required for DBE CAPS moderation.
- Read through all questions and reference notes carefully before attempting solutions.
- Answer ALL questions according to the specific rubrics and curriculum guidelines provided.
- Number the answers correctly according to the numbering system used in this question paper.
- Write neatly and legibly. Work must show all calculation steps, units, and clear final statements.

SECTION A: CORE SYLLABUS CONCEPTS & QUESTIONS (Mathematics Grade 10)

QUESTION 1: ALGEBRAIC EQUATIONS, EXPRESSIONS & FACTORISATION [25 MARKS]

- 1.1 Solve for x in the following expressions:
- | | |
|---------------------------|-----------|
| 1.1.1 $3x^2 - 5x - 2 = 0$ | (3 marks) |
| 1.1.2 $2x - 3 = (x + 6)$ | (4 marks) |
| 1.1.3 $x(x - 4) < 12$ | (4 marks) |
- 1.2 Simplify fully without the use of a calculator:
- | | |
|---|-----------|
| 1.2.1 $(2^{n+1} \cdot 4^{n-1}) / (8^n)$ | (4 marks) |
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QUESTION 2: FUNCTIONS, GRAPHS & HYPERBOLAS [30 MARKS]

- 2.1 Given the function $f(x) = \frac{2}{x-1} + 3$:
- | | |
|--|-----------|
| 2.1.1 Write down the equations of the asymptotes of f . | (2 marks) |
| 2.1.2 Calculate the coordinates of the x and y intercepts of f . | (4 marks) |
| 2.1.3 Sketch the graph of f , showing all asymptotes clearly. | (5 marks) |

AUTHENTICATED CAPS EXAMINATION MATERIAL

Produced by Fusion High School Management System • Verified by Curriculum Head: Dr. T. Makola
This document is official study material aligned with the DBE National Curriculum Statement (South Africa).